



Representing species distributions in spatially-explicit ecosystem models from presence-only data



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ABSTRACT

Spatially-explicit ecosystem models are increasingly considered for informing fisheries management. The inputs of these models that determine species distributions in the modeled system are critical. There is a need for methods to estimate species distributions in spatially-explicit ecosystem models from presence-only data. To address this need, we used a method relying on binomial generalized additive models integrating environmental covariates. This method allows for the production of distribution maps for ecosystem models such as Atlantis models, and of preference functions for Ecospace models; preference functions define the preferences of species groups for certain environmental parameter values and are employed by Ecospace to allocate species group biomasses spatially. The key step of the method we used is the objective generation of pseudo-absences for each month, by sampling with replacement the centroids of the cells of fine-scale spatial grids defined for each month. To demonstrate the method, we applied it to the diving bird and surface-feeding bird groups represented in the Atlantis model for the Gulf of Mexico (GOM), and to the seabird group represented in the Ecospace model for the West Florida Shelf. We also employed the distribution maps we constructed to provide a basis for a hypothetical marine protected area (MPA) planning scenario aiming to secure food for seabirds. Specifically, we produced a hotspot map for seabirds for the U.S. GOM from the distribution map of the species group, and then combined the hotspot map for seabirds with a hotspot map for their main prey, forage fish (Clupeidae and Exocoetidae), to determine where the hotspots of seabirds and forage fish overlap. This analysis suggested that, to secure forage fish for seabirds in the U.S. GOM, hypothetical MPAs should be implemented primarily within the coastal regions of the Louisiana-Texas shelf, within the coastal region of West Florida located between Sarasota and Naples and/or within Apalachee Bay, Florida.

1. Introduction

Spatially-explicit ecosystem models, which simulate interactions among species groups, interactions between species groups and fisheries, and species-environment relationships, are increasingly considered for informing fisheries management (Christensen and Walters, 2011; Lehuta et al., 2016; O'Farrell et al., 2017). In particular, the Atlantis (Fulton et al., 2004, 2011), OSMOSE (Shin and Cury, 2001; Grüss et al., 2016b) and Ecospace (Walters et al., 1999, 2010) modeling platforms are increasingly being employed worldwide, including in the

Gulf of Mexico (GOM; Fig. 1). Applications of the Atlantis and Ecospace modeling approaches usually represent a comprehensive suite of species groups from primary and secondary producers to top predators (large predatory fish, marine mammals, sea turtles, and seabirds). This level of sophistication entails that Atlantis and Ecospace be provided with many inputs (Fulton et al., 2007; Shin et al., 2010; Walters et al., 2010). However, the many inputs fed into Atlantis and Ecospace models need to be of reasonable quality if the predictions made by these models are to be used to inform fisheries management decisions (Drexler and Ainsworth, 2013; Grüss et al., 2016a; Sagarese et al., 2016; Tarnecki

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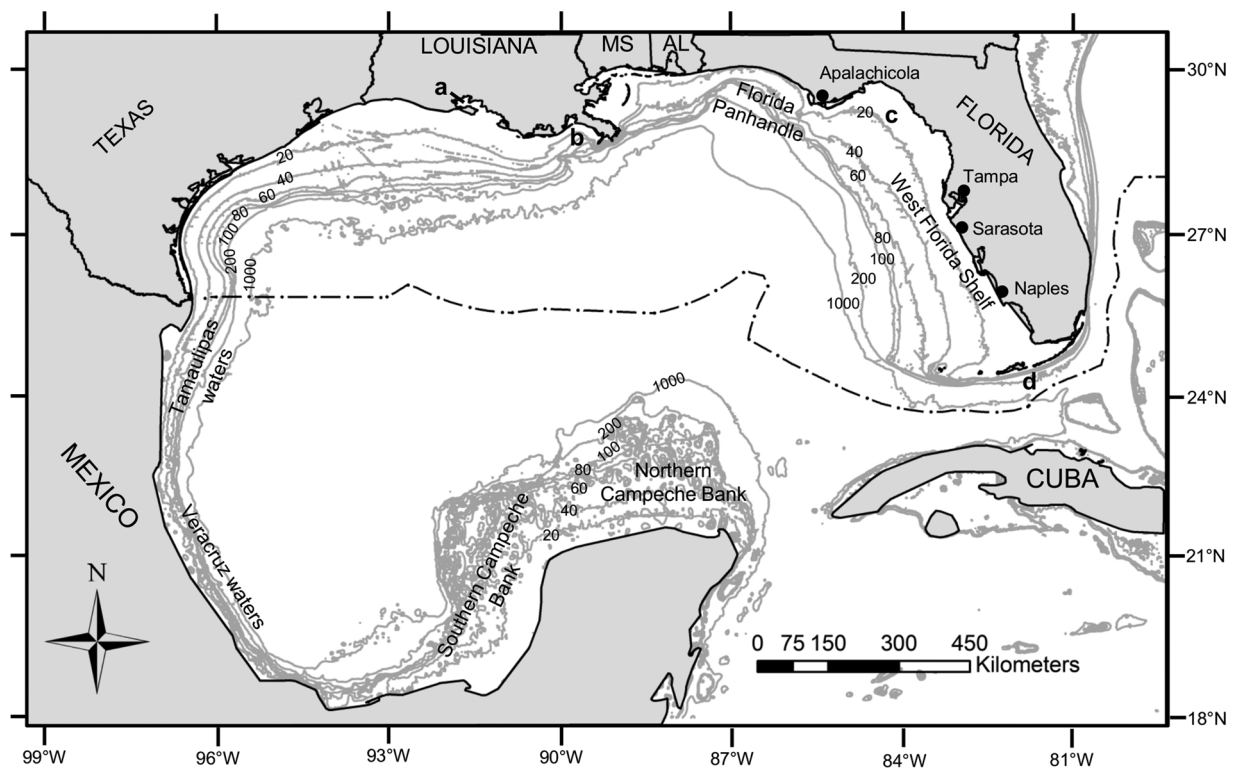


Fig. 1. Map of the Gulf of Mexico Large Marine Ecosystem. Depth contours are labeled in 20 – , 40 – , 60 – , 80 – , 100 – , 200 – , and 1000 – m contours. Important features are labeled and include: northern Campeche Bank, southern Campeche Bank, Veracruz waters, Tamaulipas waters, Vermilion Bay (a), Barataria Bay (b), the Florida Panhandle, the West Florida Shelf, Apalachee Bay (c), and the Florida Keys (d). MS = Mississippi - AL = Alabama. The black dashed-dotted line delineates the U.S. exclusive economic zone.

et al., 2016).

Some of the critical inputs to spatially-explicit ecosystem models are those that determine the spatial distribution of species groups, since those inputs govern the degree of spatial overlap among species groups and between species groups and fisheries and environmental events, which, in turn, can affect animal mortality, growth and reproduction processes (Grüss et al., 2016a). To allocate species groups spatially, applications of the Atlantis and OSMOSE modeling platforms need to be provided with maps defining the annual or seasonal distribution patterns of species groups in the horizontal dimension (Drexler and Ainsworth, 2013; Grüss et al., 2014). Ecospace models do not require distribution maps for allocating species group biomasses over space, but rather preference functions, which describe the preferences of species groups for specific environmental conditions (Christensen et al., 2014). At run time, Ecospace combines these preference functions with spatio-temporal environmental data in a calculation of species group spatial distributions using the “habitat capacity model” module (Steenbeek et al., 2013).

Until recently, the production of distribution maps and preference functions for spatially-explicit ecosystem models has relied on simplistic methods (see Grüss et al. (2016a) for a review). To remedy this issue and ensure that spatially-explicit ecosystem models are provided with more reasonable inputs for allocating species group biomasses spatially, a large statistical modeling project was conducted in the GOM. This involved compiling the monitoring data for the U.S. (northern) GOM for fish and invertebrates collected using random sampling schemes, fitting species distribution models (SDMs) to presence/absence data, and employing SDM predictions to generate inputs for spatially-explicit ecosystem models for the GOM (Grüss et al., 2018a,b,d). Grüss et al. (2018d) fitted geostatistical binomial generalized linear mixed models (GLMMs) to presence/absence data from the large monitoring database to construct distribution maps for the OSMOSE model of the West Florida Shelf. Grüss et al. (2018a) fitted

binomial generalized additive models (GAMs) integrating environmental covariates to presence/absence data from the large monitoring database to produce preference functions for the Ecospace model of the West Florida Shelf (referred to as “WFS Reef fish Ecospace”). Finally, Grüss et al. (2018b) used a hybrid approach and combined the predictions of geostatistical binomial GLMMs for the U.S. GOM and the predictions of binomial GAMs integrating environmental covariates for the southern GOM to generate distribution maps for the Atlantis model of the GOM (referred to as “Atlantis-GOM”).

While the above-mentioned studies led to the production of a large number of inputs for spatially-explicit ecosystem models for the GOM, they focused on fish and invertebrates, for which presence/absence data are available. However, sightings (i.e., presence-only data) rather than presence/absence data are available for many species and species groups represented in spatially-explicit ecosystem models (Pearce and Boyce, 2006; Aarts et al., 2012; Renner et al., 2015). Thus, in the GOM, presence-only data for West Indian manatee (*Trichechus manatus latirostris*), a sirenian species represented in the Atlantis-GOM model, are collected using an ensemble of aerial surveys in the northeastern GOM (Martin et al., 2015), and mainly through a citizen science program in the northwestern GOM (Carmichael et al., 2017; Hieb et al., 2017). Another example in the GOM is that of the seabird groups represented in Atlantis-GOM and WFS Reef fish Ecospace, for which presence-only data have been collected in the U.S. GOM by a handful of monitoring programs with different spatio-temporal resolutions (Davis et al., 2000; Ford, 2010; Love et al., 2015; DIVER, 2017).

Several approaches are available for developing SDMs using presence-only data. The most well-known of those is the MaxEnt approach, which relies on an environmental landscape that defines the value of environmental parameter values over a spatial grid (Phillips et al., 2006, 2017). Using this environmental landscape, MaxEnt generates pseudo-absence locations, which are then contrasted against the locations of the presence data to estimate species spatial distributions

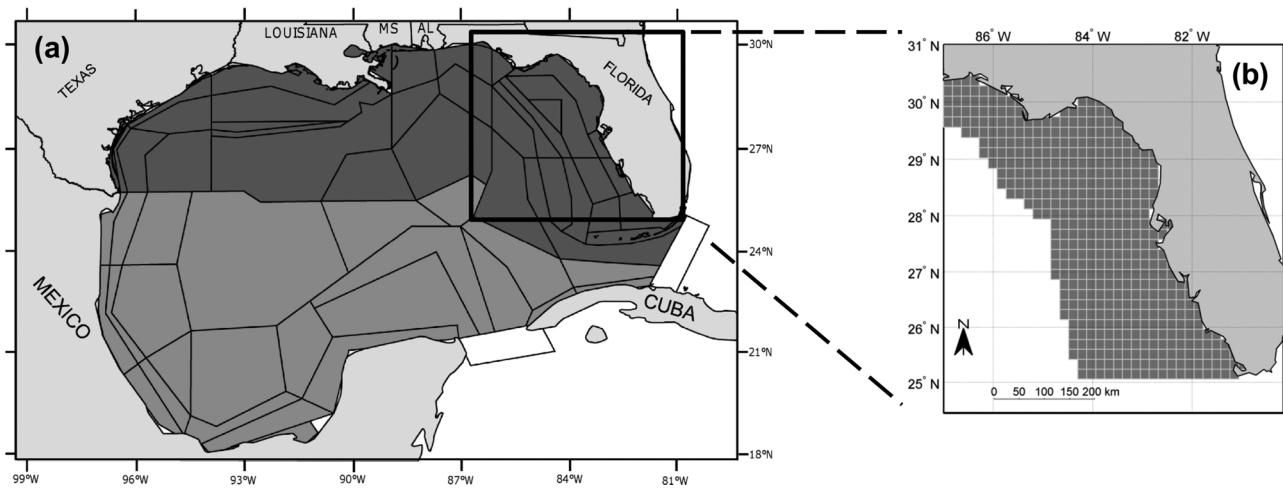


Fig. 2. Spatial domains of the ecosystem models considered in this study. (a) Spatial polygons of the Atlantis model of the Gulf of Mexico (GOM) referred to as “Atlantis-GOM”. Atlantis-GOM polygons located in U.S. GOM are filled in dark gray, while those located outside the U.S. GOM (i.e., in the Mexican or Cuban GOM or in international waters) are filled in lighter gray. The two “boundary polygons” of Atlantis-GOM, which do not interact with the rest of the model, are filled in white. MS = Mississippi - AL = Alabama. (b) Spatial cells of the Ecospace model of the West Florida Shelf referred to as “WFS Reef fish Ecospace” (filled in dark grey).

(Merow et al., 2013; Phillips et al., 2017). Another frequently-employed method is the Poisson point process modeling approach, which has a number of advantages over the MaxEnt approach (Warton and Shepherd, 2010; Renner and Warton, 2013; Renner et al., 2015). In particular, contrary to the MaxEnt approach, the Poisson point process modeling approach allows for an objective definition of the number and location of pseudo-absences; more precisely, with Poisson point process models, pseudo-absences are selected within the spatial extent of the region of interest either at random or on a regular mesh (Renner and Warton, 2013; Renner et al., 2015).

To fill gaps in the provision of distribution maps and preference functions to spatially-explicit ecosystem models, we built upon existing approaches that deal with presence-only data and used a method relying on presence-only data and binomial GAMs integrating environmental covariates. We chose to employ a GAM framework rather than either MaxEnt or Poisson point process models for several reasons. First, we preferred having control over the generation of pseudo-absences rather than letting the MaxEnt approach deal entirely with this procedure (Renner et al., 2015). Second, we needed to fit SDMs to monitoring data for the U.S. GOM and then perform extrapolation with the SDMs over the entire GOM large marine ecosystem (LME) to obtain distribution maps for the Atlantis-GOM model; in this context, we wanted to rely on flexible models such as GAMs with thin-plate regression splines restricted to a limited number of degrees of freedom (Drexler and Ainsworth, 2013; Grüss et al., 2014; Mannocci et al., 2017). Moreover, preference functions constructed from the predictions of GAMs with thin-plate regression splines restricted to a limited number of degrees of freedom are much easier to interpret than preference functions constructed from the complex partial dependence plots derived from MaxEnt models (Phillips et al., 2006; Roberts et al., 2016; Mannocci et al., 2017).

We provide an illustration of our work for the seabird groups represented in the Atlantis-GOM and WFS Reef fish Ecospace ecosystem models. This illustration can serve future projects aiming to generate distribution maps and preference functions for spatially-explicit ecosystem models of the GOM, and of other marine regions, from presence-only data. In the following, we first briefly present the study regions (the GOM LME and the West Florida Shelf), the study ecosystem models (Atlantis-GOM and WFS Reef fish Ecospace) and the study species groups. Then, we provide an overview of the method we used. Next, we describe the seabird monitoring data and the environmental data we selected for integration into GAMs. Afterwards, we explain how we developed and evaluated GAMs for the seabird groups represented in

Atlantis-GOM and WFS Reef fish Ecospace, and how we used GAM predictions to construct distribution maps and preference functions for Atlantis-GOM and WFS Reef fish Ecospace. Finally, to illustrate the usefulness of the products generated using our approach beyond the provision of inputs to spatially-explicit ecosystem models, we employ the distribution maps we produced to provide a basis for a hypothetical marine protected area (MPA) planning scenario. More specifically, we use the map generated for seabirds (all species combined) and a map generated for their main prey, Clupeidae and Exocoetidae (henceforth “forage fish”), to determine the areas of the U.S. GOM where the hotspots of seabirds and those of forage fish overlap, to then provide a basis for a hypothetical MPA planning scenario aiming to secure food for seabirds.

2. Material and methods

2.1. Study regions, ecosystem models and species groups

The GOM is a LME bordered by three countries: the U.S., Cuba, and Mexico (Fig. 1). The first ecosystem model considered in this study is Atlantis-GOM (Ainsworth et al., 2015, 2018), whose spatial domain covers the entire GOM LME and comprises 63 irregular polygons, as well as two “boundary polygons” which do not interact with the rest of model (Fig. 2a). In Atlantis-GOM, 91 species groups are represented, including two seabird groups: diving birds and surface-feeding birds. The diving bird group consists of 24 species and the surface-feeding bird group includes six species (Table 1).

The second ecosystem model is WFS Reef fish Ecospace (Chagaris, 2013), whose spatial domain covers the entire West Florida Shelf region excluding the Florida Keys and comprises 465 regular 0.18° ($20\text{ km} \times 20\text{ km}$) cells (Fig. 2b). Seventy-one species groups are represented in WFS Reef fish Ecospace, including one seabird group. The seabird group represented in WFS Reef fish Ecospace (hereafter generally simply referred to as “seabirds”) comprises all the seabird species represented in Atlantis-GOM (30 species in total; Table 1). Thus, three species groups were considered: diving birds, surface-feeding birds, and seabirds.

2.2. Overview of the method used in this study

There are seven steps to produce distribution maps and preference functions (Fig. 3). First, we compiled a dataset gathering the seabird sightings (presence-only data) collected in the U.S. GOM retained for this study, as well as an environmental database for the GOM LME

Table 1

Seabird groups and species represented in the Atlantis-GOM ecosystem model. The single seabird group represented in the WFS Reef fish Ecospace ecosystem model comprises all the species making up both the diving bird and surface-feeding bird groups represented in Atlantis-GOM.

Seabird group	Species making up the group
Diving birds	Brown pelican (<i>Pelecanus occidentalis</i>), double-crested cormorant (<i>Phalacrocorax auritus</i>), masked booby (<i>Sula dactylatra</i>), brown booby (<i>Sula leucogaster</i>), northern gannet (<i>Morus bassanus</i>), common loon (<i>Gavia immer</i>)
Surface-feeding birds	Wilson's storm petrel (<i>Oceanites oceanicus</i>), black tern (<i>Chilodonia niger</i>), herring gull (<i>Larus argentatus</i>), laughing gull (<i>Larus atricilla</i>), ring-billed gull (<i>Larus delawarensis</i>), black skimmer (<i>Rynchops niger</i>), parasitic jaeger (<i>Stercorarius parasiticus</i>), pomarine jaeger (<i>Stercorarius pomarinus</i>), least tern (<i>Sternula antillarum</i>), caspian tern (<i>Sterna caspia</i>), Foster's tern (<i>Sterna forsteri</i>), common tern (<i>Sterna hirundo</i>), royal tern (<i>Sterna maxima</i>), gull-billed tern (<i>Sterna nilotica</i>), sandwich tern (<i>Sterna sandvicensis</i>), American white pelican (<i>Pelecanus erythrorhynchos</i>), osprey (<i>Pandion haliaetus</i>), magnificent frigatebird (<i>Fregata magnificens</i>), Audubon's shearwater (<i>Puffinus lherminieri</i>), great shearwater (<i>Puffinus gravis</i>), Cory's shearwater (<i>Calonectris diomedea</i>), Leach's storm petrel (<i>Oceanodroma leucorhoa</i>), band-rumped storm petrel (<i>Oceanodroma castro</i>), bridled tern (<i>Onychoprion anaethetus</i>)

gathering the environmental variables known to influence the spatial distribution patterns of most seabird species that are available for the entire GOM LME. Second, we generated pseudo-absences for each month; we hence obtained presence/pseudo-absence data for each month. Third, we assessed the degree of collinearity between the environmental predictors considered in this study, as well as between these environmental predictors and eastings and northings (i.e., longitude and latitude expressed in UTM coordinates), resulting in the selection of a subset of the original environmental predictors for statistical analyses. Fourth, we fitted binomial GAMs to presence/pseudo-absence data and selected environmental predictors. Fifth, we evaluated the fitted binomial GAMs using an iterative cross-validation procedure. Sixth, we made predictions with the fitted binomial GAMs, to then produce distribution maps with a resolution of 0.18° (i.e., with a 20 km × 20 km resolution), as well as preference functions for the WFS Reef fish Ecospace ecosystem model. Lastly, in the cases of diving birds and surface-feeding birds, we constructed distribution maps for the Atlantis-GOM ecosystem model from their 0.18° distribution maps. The 0.18° (20 km × 20 km) resolution of the distribution maps produced in the sixth step was chosen, because this is the resolution of the WFS Reef fish Ecospace ecosystem model.

2.3. Preparation of a seabird monitoring dataset

We contacted the state and federal agencies, academic researchers, and non-governmental organizations that monitor seabirds in the U.S. GOM and requested access to their data. The monitoring data we received had various spatio-temporal coverages. In particular, several monitoring datasets had an extremely limited spatio-temporal resolution. Ultimately, we selected the DIVER (Data Integration, Visualization, Exploration, and Reporting) low-level aerial bird survey for this study (Ford, 2010).

This dataset is part of the larger DIVER dataset compiled by the National Oceanic and Atmospheric Administration (NOAA) (DIVER, 2017). DIVER is a data repository for the integration and distribution of the Natural Resource Damage Assessment (NRDA) data collected after the Deepwater Horizon oil spill of 2010. The observations were collected across the entire U.S. GOM between May 2010 and February 2011 (Fig. 4a). The survey employed standard aerial seabird survey protocols (Briggs et al., 1985; Henkel et al., 2007). These protocols consisted of zigzag transects between the coastline and barrier islands, as well as transects conducted offshore up to 100 nautical miles seaward (Fig. 4a). Surveys were flown at a speed of 90–100 knots and at an altitude of ca. 200 feet above sea level (Ford, 2010). They involved a pilot, a navigator/data recorder, as well as two observers who were positioned on each side of the rear of the aircraft. Time, latitude and longitude were recorded via a GPS unit linked to an onboard computer (Ford, 2010). While “on effort”, if the observers observed seabirds, they recorded information (species/species types observed, estimated counts, and behavior) on a digital voice recorder. This information was then relayed via the aircraft intercom to the navigator/data recorder, who merged it with the GPS information (Ford, 2010).

2.4. Generation of pseudo-absences for seabird groups

The survey provided presence-only data. However, to be able to fit binomial GAMs for seabird groups, we needed presence/pseudo-absence data. We, therefore, generated pseudo-absences for each of the three species groups by selecting pseudo-absences at random within the regions covered by the DIVER low-level aerial bird survey. Thus, we obtained presence and pseudo-absence data that were random samples from our study geographical area. As mentioned earlier, the selection of pseudo-absences within the spatial extent of the region of interest at random is one way pseudo-absence locations are defined in Poisson point process modeling studies (Renner and Warton, 2013; Renner et al., 2015), which we built upon in the present study. This approach for generating pseudo-absences is recommended in other studies as well (Pearce and Boyce, 2006; Aarts et al., 2012).

To be able to obtain presence/pseudo-absence data for each month, we first created a fine-scale (0.01°) spatial grid for each month. Poisson point process modeling studies, which we built upon in the present study, consider grids of increasing resolution for generating pseudo-absences that allow estimation of the model likelihood (Warton and Shepherd, 2010; Renner et al., 2015). Hence, we produced spatial grids for generating pseudo-absences that had a resolution much finer than that of our prediction grids and distribution maps (0.18°). To create spatial grids for each month, we first constructed a 0.01° spatial grid covering the entire U.S. GOM. We then employed the Gulf of Mexico Coastal Observing System GCOOS bathymetry data, which are a derivative of the original SRTM30 PLUS global bathymetry data (Becker et al., 2009) to generate a depth raster for the U.S. GOM with a resolution of 0.01°. Next, we used the generated depth raster to estimate the depth ranges of the areas of the U.S. GOM where seabirds were observed by the DIVER low-level aerial bird survey. Finally, we created the spatial grids for each month, based on the ranges of latitude, longitude and depth where seabirds were observed by the DIVER low-level aerial bird survey (Grüss et al., 2018d).

We employed the 0.01° spatial grids to generate pseudo-absences. This involved sampling the centroids of the cells of the 0.01° spatial grids constructed for each month with replacement (Pearce and Boyce, 2006; Aarts et al., 2012; Renner et al., 2015). The log-likelihood of Poisson point process modeling studies (Warton and Shepherd, 2010; Renner et al., 2015), which we built upon in the present study, converged when at least one order of magnitude more pseudo-absences than presences were produced. Hence, for each month, we decided to generate ten times more pseudo-absences than the number of sightings. The R code for generating pseudo-absences is provided in Appendix A1.

2.5. Construction of an environmental database

We conducted a literature review to identify the environmental parameters influencing the spatial distribution patterns of the seabird groups represented in Atlantis-GOM and WFS Reef fish Ecospace and for which data are available for the entire GOM LME. Following primarily Ribic et al. (1997), Tremblay et al. (2009) and Poli et al. (2017),

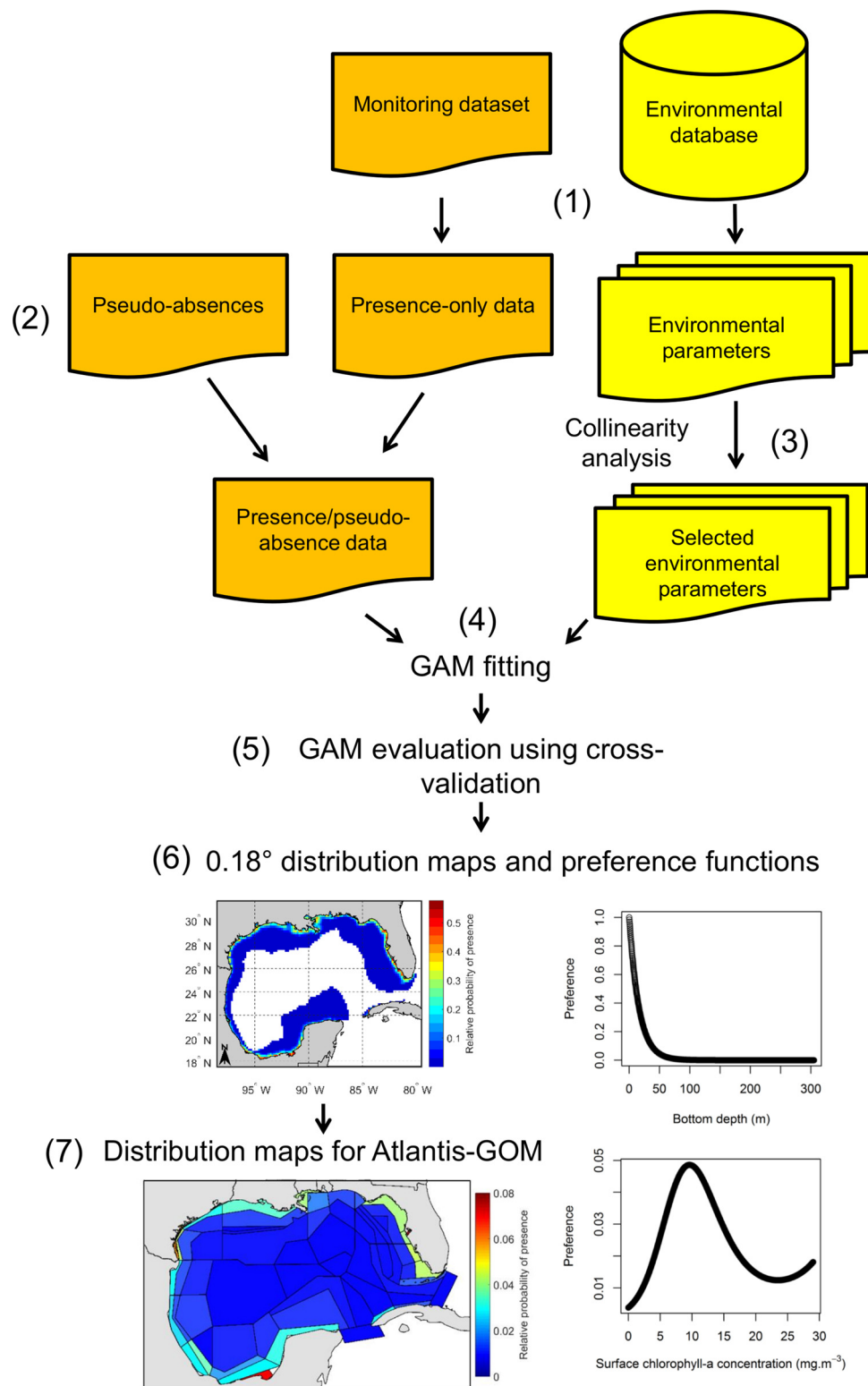


Fig. 3. Schematic of the steps followed to produce distribution maps and preference functions in this study.

and based on data availability, we integrated the following parameters into an environmental database: (1) bottom depth; (2) distance from shore; (3) sea surface temperature (SST); (4) precipitation; (5) surface chlorophyll-a concentration; (6) oceanic current speed; (7) wind speed; and (8) sea surface height (SSH) (Table 2). Bottom depth and distance from shore are constant over time, while the other environmental parameters were defined for each of the months of the year via the

production of “climatologies” describing monthly conditions during the period 2000–2015 (Table 2).

To be able to fit GAMs and make predictions from the fitted GAMs, we constructed a gridded map of environmental parameters for the GOM LME with a resolution of 0.18° (i.e., the resolution of the WFS Reef fish Ecospace ecosystem model) from the environmental database (Fig. A2). This gridded map was overlaid with presence/pseudo-

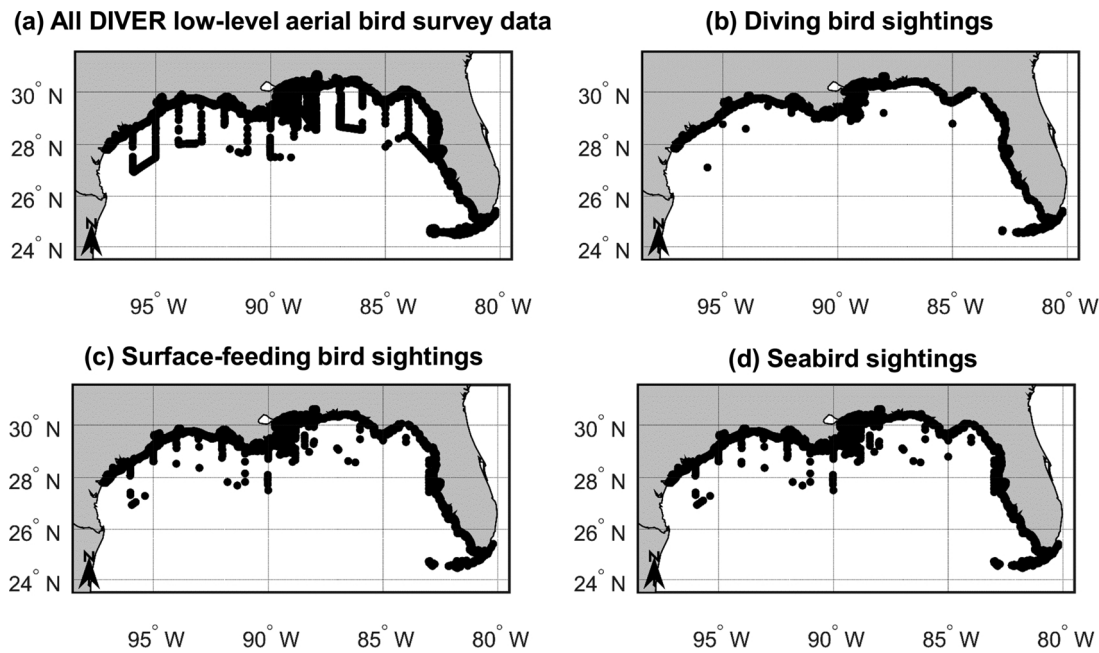


Fig. 4. Spatial distributions of the seabird sightings collected by the DIVER (Data Integration, Visualization, Exploration, and Reporting) low-level aerial bird survey in the U.S. Gulf of Mexico. (a) is for all seabird species combined, including those not considered in this study, (b) is for species of the diving bird group represented in the Atlantis-GOM ecosystem model, (c) is for the species of the surface-feeding bird group represented in Atlantis-GOM, and (d) is for species of the seabird group represented in the WFS Reef Ecospace ecosystem model. The compositions of the diving and surface-feeding bird groups represented in Atlantis-GOM and of the seabird group represented in WFS Reef fish Ecospace are given in Table 1.

Table 2

Data used to generate spatial estimates for the environmental parameters integrated in the generalized additive models (GAMs) developed in this study, and manipulations required on these data to attain a continuous surface with which to fit GAMs and make predictions with fitted GAMs.

Environmental parameter	Data used and data manipulations for GAMs	Range values in the West Florida Shelf region
Bottom depth	We accessed the Gulf of Mexico Coastal Observing System bathymetry data, which is a derivative of the original SRTM30 PLUS global bathymetry data (Becker et al., 2009), from which a continuous raster of bathymetry for the GOM large marine ecosystem (LME) with a resolution of 0.18° was produced.	0 – 305 m
Distance from shore	Distance from shore was estimated for the entire GOM LME in MATLAB using a dedicated function ("dist_from_coast"), which can be accessed here: https://www.mathworks.com/matlabcentral/fileexchange/47632-dist_from_coast-zip?focused=3835717&tab=function	0 – 262 km
Sea surface temperature	Aqua MODIS monthly 0.0125-degree sea surface temperature composites (daytime; 11 microns) for the period 2002–2011 were downloaded (Simons, 2017), from which we estimated sea surface temperature in each of the cells of a 0.18° grid covering the entire GOM LME for the different months of the year.	14 – 32 °C
Precipitation	PERSIANN-CDR 0.25-degree resolution daily precipitation data for the period 2000–2015 (Sorooshian et al., 2014; Ashouri et al., 2015) were downloaded, from which we estimated precipitation in each of the cells of 0.18° grid covering the entire GOM LME for the different months of the year.	30 – 270 mm
Surface chlorophyll-a concentration	Aqua MODIS monthly 0.0125-degree surface chlorophyll-a (chl-a) concentration composites for the period 2002–2012 (Simons, 2017), from which we estimated surface chl-a concentration in each of the cells of a 0.18° grid covering the entire GOM LME for the different months of the year.	0 – 29 mg.cm ⁻³
Oceanic current speed	OSCAR (Ocean Surface Current Analyses Real-time) third degree resolution ocean surface current data for the period 2005–2015 (ESR, 2009) were downloaded, from which we estimated oceanic current speed in each of the cells of a 0.18° grid covering the entire GOM LME for the different months of the year.	0 – 0.51 m.s ⁻¹
Wind speed	NOAA/NCDC blended monthly 0.25-degree sea surface wind composites for the period 2000–2011 were downloaded from http://coastwatch.pfeg.noaa.gov/erddap/griddap/ , from which we estimated wind speed in each of the cells of a 0.18° grid covering the entire GOM LME for the different months of the year.	4 – 8.5 m.s ⁻¹
Sea surface height	Aviso monthly 0.25-degree sea surface height composites for the period 2000–2009 were downloaded from http://coastwatch.pfeg.noaa.gov/erddap/griddap/ , from which we estimated sea surface height in each of the cells of a 0.18° grid covering the entire GOM LME for the different months of the year.	1.1 – 1.7 m

absence data to fit GAMs. It was also overlaid on a 0.18° spatial grid covering the entire GOM LME to enable us to predict relative probabilities of presence over the entire U.S. GOM or over the entire GOM LME to then generate distribution maps and preference functions for ecosystem models.

2.6. Collinearity analysis

Regression modeling may be sensitive to correlated predictors (Guisan et al., 2002; Dormann et al., 2013). For this reason, for each species group (diving birds, surface-feeding birds, and seabirds), before fitting its GAM, we assessed the degree of collinearity between the

environmental predictors (i.e., bottom depth, distance from shore, SST, precipitation, surface chlorophyll-*a* concentration, oceanic current speed, wind speed, and SSH), as well as between each of these environmental predictors and eastings and northings in the case of seabirds only. More specifically, we estimated Pearson's correlation coefficients between the environmental predictors for each of the species group, as well as Pearson's correlation coefficients between each of the environmental predictors and eastings and northings in the case of seabirds only (Booth et al., 1994; Brotons et al., 2004; Dormann et al., 2013). Then, for each species group, we removed one of the two environmental predictors of the pair if we observed absolute pairwise correlations greater than 0.7 (Leathwick et al., 2006; Dormann et al., 2013; Grüss et al., 2018a).

2.7. GAM fitting

After the collinearity analysis was conducted, we fitted two types of binomial GAMs: (1) “non-spatial GAMs”, which integrated only environmental predictors; and (2) “spatial GAMs”, which integrated environmental predictors, as well as an interaction term between eastings and northings that accounted for spatial structure (spatial autocorrelation) at a broad scale (Wood, 2006; Grüss et al., 2018a). Non-spatial GAMs were developed for diving birds and surface-feeding birds, and a spatial GAM was developed for the seabird group represented in WFS Reef fish Ecospace. Thus, we developed three binomial GAMs in this study. Since the survey recorded estimated counts every time seabirds were sighted, we could have developed Poisson GAMs or delta-Poisson GAMs (e.g., Barry and Welsh, 2002; Grüss et al., 2014) in lieu of binomial GAMs. However, since our objective was to predict spatial patterns of relative probability of presence for seabird groups of the GOM, it was more relevant to develop binomial GAMs in the present study.

All binomial GAMs were fitted to the presence/pseudo-absence data using the R package “mgcv” (Wood and Augustin, 2002; Wood, 2006). Non-spatial binomial GAMs were fitted following the equation:

$$g(\eta) = s(x_1) + s(x_2) + \dots + s(x_n) \quad (1)$$

where η is the relative probability of presence; g is the logit link function between η and each additive predictor; s is a thin plate regression spline fitted to a given environmental predictor; and $x_1, x_2, \dots, x_n$ are the environmental predictors selected after the collinearity analysis.

The spatial binomial GAM of seabirds was similar to the non-spatial GAMs, except that it also integrated a tensor product smooth fitted to eastings and northings ($te(X, Y)$):

$$g(\eta) = te(X, Y) + s(x_1) + s(x_2) + \dots + s(x_n) \quad (2)$$

We utilized thin-plate regression splines with shrinkage, and we restricted each thin-plate regression spline to four degrees of freedom (Roberts et al., 2016; Mannocci et al., 2017). This allowed us to maintain the interpretability of the relationships estimated by GAMs and, consequently, of the preference functions derived from GAM predictions (Grüss et al., 2018a). We used a shrinkage approach for selecting environmental predictors for the GAMs (Roberts et al., 2016; Mannocci et al., 2017). In addition, we applied an extra penalty to each environmental predictor, which allowed for the complete removal of an environmental predictor when the smoothing parameter was zero (Marra and Wood, 2011). As was done in many previous studies (e.g., Koubbi et al., 2006; Weber and McClatchie, 2010; Grüss et al., 2014, 2016c, 2018a,b; Chagaris et al., 2015), after a GAM was fitted, if an environmental parameter was estimated to have a p -value greater than 0.05, this parameter was dropped and the GAM was refitted; we followed this procedure to obtain the simplest possible GAMs integrating only relevant environmental predictors. As previous studies (e.g., Roberts et al., 2016; Mannocci et al., 2017; Grüss et al., 2018a,b), we employed the restricted maximum likelihood (REML) optimization method (Wood, 2011) for estimating the smoothing parameters of

GAMs that do not include any random effects. We did so, because it was shown that REML penalizes overfitting and generally results in better mean-square error performance than the other optimization methods available with the R package “mgcv” (Wood, 2011). The code for fitting GAMs is provided in Appendix A3.

Typically, ecological data are spatially autocorrelated (Legendre, 1993; Dormann et al., 2007; Renner et al., 2015). The non-spatial GAMs of diving birds and surface-feeding birds do not account for spatial structure, because those GAMs were developed for performing extrapolation over the entire GOM LME, including the southern GOM where seabird monitoring data were not available (Drexler and Ainsworth, 2013; Mannocci et al., 2017). We may, therefore, expect the presence of significant, unmodeled spatial patterns in the residuals of non-spatial GAMs. By contrast, the spatial GAM of seabirds accounts for spatial structure at a broad scale through the inclusion of the $te(X, Y)$ term (Eq. (2)). The main rationale behind the integration of the $te(X, Y)$ term in the spatial GAM of seabirds was to enable the statistical interpretation of significance for environmental covariates when GAM residuals have spatial patterns (Wood, 2006; Grüss et al., 2018a). Yet, despite the integration of the $te(X, Y)$ term, non-negligible spatial patterns may be present in the residuals of the spatial GAM of seabirds. To characterize spatial structure in the residuals of the non-spatial and spatial GAMs developed in this study, we constructed empirical variograms of GAM residuals (Appendix A4). An empirical variogram with a pattern indicates the presence of spatial structure, while one that is primarily flat reveals an absence of spatial autocorrelation in the data (Cressie, 1993). Typically, the empirical variograms produced for ecological data have a roughly logarithmic form. However, the presence of spatial autocorrelation can also be revealed by the existence of spatial clusters of positive and negative GAM residuals (Cressie, 1993; Grüss et al., 2016c).

2.8. GAM evaluation

To evaluate the final GAM obtained for each species group (i.e., after some environmental parameters were possibly dropped based on their p -value), we carried out a procedure similar to the “Leave Group Out Cross Validation” procedure implemented in the R package “dismo” (Hijmans et al., 2015). In this procedure, presence/pseudo-absence data were randomly split into training and test datasets, and 60% of the presence/pseudo-absence data were employed for GAM training and the rest for GAM evaluation. For each species group, within the evaluation process, GAM training consisted of using the training dataset to re-fit the final GAM obtained for the species group under consideration. Then, the test dataset corresponding to the training dataset was employed to make predictions with the GAM resulting from the GAM training step. We repeated this process 10 times for each species group (Grüss et al., 2016c).

For each species group, two diagnostics were utilized to evaluate binomial GAMs through the Leave Group Out Cross Validation procedure: (1) the area under the receiver operating characteristic (ROC) curve (AUC), which expresses the chance that a presence estimate chosen randomly has a larger predicted relative probability of presence than a pseudo-absence estimate chosen randomly (Hanley and McNeil, 1982; van Proosdij et al., 2016); and (2) calibration plots, which assess the ability of models to correctly predict the proportion of locations with the species group under consideration present, such that perfect calibration corresponds to a line with a slope equal to 1 and an intercept equal to 0 (Wintle et al., 2005; Heinänen et al., 2008; Sagarese et al., 2014). Following Hanley and McNeil (1982); Swets (1988) and Pearce and Ferrier (2000), each of the binomial GAMs we fitted was validated if its median AUC value estimated via the Leave Group Out Cross Validation procedure was higher than 0.7. The maximum value that AUC can achieve is smaller than 1 and is usually unknown when applied to presence/pseudo-absence data; for this reason, several authors (e.g., Lobo et al., 2008; Jiménez-Valverde, 2012; van Proosdij et al., 2016)

have recommended not using AUC as the only diagnostic for GAM evaluation when working with presence/pseudo-absence data. Hence, we considered both the AUC metric and calibration plots to evaluate the GAM of each species group. The code for GAM evaluation is provided in Appendix A3.

2.9. Production of distribution maps from GAM predictions

Predictions were made with the fitted and validated GAMs to produce three types of distribution maps: (1) a map of relative probability of presence in the U.S. GOM for seabirds, with a resolution of 0.18° (to be used to provide a basis for a hypothetical MPA planning scenario; see below); (2) maps of relative probability of presence in the GOM LME for diving birds and surface-feeding birds, with a resolution of 0.18° ; and (3) distribution maps usable in Atlantis-GOM for diving birds and surface-feeding birds, generated from their maps of relative probability of presence in the GOM LME.

Before producing the three types of distribution maps, 0.18° prediction grids were constructed for seabirds, diving birds and surface-feeding birds. The construction of predictions grids started with the creation of 0.18° spatial grids covering the entire U.S. GOM and the entire GOM LME. We then employed the GCOOS bathymetry data, which is a derivative of the original SRTM30 PLUS global bathymetry data (Becker et al., 2009) to generate a depth raster for the U.S. GOM with a resolution of 0.18° . Next, we used the generated depth raster to estimate the depth ranges of the areas of the U.S. GOM where seabirds, diving birds, and surface-feeding birds were observed by the survey. Finally, we constructed a 0.18° prediction grid for the U.S. GOM for seabirds and 0.18° prediction grids for the GOM LME for diving birds and surface-feeding birds, based on the estimated depth ranges at which each of the three species groups were observed by the survey in the U.S. GOM (Grüss et al., 2018b).

After the prediction grids were constructed, we predicted the spatial patterns of relative probability of presence of seabirds in the U.S. GOM and of diving birds and surface-feeding birds in the GOM LME, using fitted GAMs, the prediction grids, and the gridded map of environmental parameters. To facilitate the interpretation of the maps of relative probability of presence of diving birds and surface-feeding birds, we assessed the relative importance of predictors in their relative probability of presence and plotted the fits of their GAMs. To assess the relative importance of predictors in relative probability of presence, we implemented the approach developed for the niche distribution modeling approach BIOMOD (Thuiller et al., 2012), which involves comparing the predictions of the binomial GAMs fitted previously (the “full GAMs”) with the predictions of binomial GAMs after random permutation of the values of a given predictor within the complete presence/pseudo-absence dataset (“random GAMs”). One minus the Pearson’s correlation coefficient between the predictions of the full GAM and the predictions of a random GAM provides an indication of the relative importance of a given predictor in explaining the relative probability of presence of diving birds or surface-feeding birds (Thuiller et al., 2012).

Finally, we employed the maps of relative probability of presence of diving birds and surface-feeding birds to generate distribution maps for the Atlantis-GOM ecosystem model. More specifically, we averaged the relative probabilities of presence of diving birds and surface-feeding birds from their 0.18° distribution maps over the extent of each Atlantis-GOM polygon and then rescaled the resulting estimates such that their sum was equal to 1.0; rescaling estimates was necessary for our products to be usable in Atlantis-GOM. The Atlantis-GOM model will utilize the rescaled relative probabilities of presence produced in this study to allocate the total biomass of diving birds and surface-feeding birds over space at the start of simulations (Grüss et al., 2018b).

2.10. Production of preference functions from GAM predictions

To construct preference functions for seabirds for WFS Reef fish

Ecospace from the spatial GAM fitted for the species group, we used the method developed in Grüss et al. (2018a). In brief, Grüss et al. (2018a)’s method works in two steps. First, for each individual environmental predictor i integrated in a GAM, predictions are made with the GAM over a vector of values ranging between the minimum and maximum values of environmental predictor i in the West Florida Shelf region (Table 2), while: (1) keeping the other environmental predictors constant at their mean value from the GAM modeled dataset; and (2) setting eastings and northings to their values at the barycenter of the WFS Reef fish Ecospace ecosystem model (27.65°N , 83.90°W). Let us denote the j th value of the i th environmental variable by $x_{i,j}$, such that $x_{i,j}$ ranges between the minimum and maximum values of the i th environmental variable in the West Florida Shelf region (Table 2), and let us denote by $p_{i,j}$ the relative probability of presence resulting from the computations made in the first step of Grüss et al. (2018a)’s method when the i th environmental variable is set to $x_{i,j}$. Then, for each individual environmental predictor, where $i \in 1, \dots, n$, the second step of Grüss et al. (2018a)’s method calculates preferences $f_{i,j}$ from $p_{i,j}$ values as follows:

$$f_{i,j} = \frac{p_{i,j} - \min\{p_{1,\min}, \dots, p_{n,\min}\}}{\max\{p_{1,\max}, \dots, p_{n,\max}\} - \min\{p_{1,\min}, \dots, p_{n,\min}\}} \quad (3)$$

where $p_{1,\min}, \dots, p_{n,\min}$ are the minimum values of relative probability of presence calculated in the first step of Grüss et al. (2018a)’s method for environmental parameters $1, \dots, n$; and $p_{1,\max}, \dots, p_{n,\max}$ are the maximum values of relative probability of presence calculated in the first step of Grüss et al. (2018a)’s method for environmental parameters $1, \dots, n$. This specification ensures that all preferences to range between 0 and 1, while accounting for the relative effect of each environmental predictor on relative probability of presence (Grüss et al., 2018a,c).

2.11. Providing a basis for a hypothetical MPA planning scenario from GAM predictions

We provided a basis for a hypothetical MPA planning scenario to protect forage fish resources used by seabirds (all species combined) in the U.S. GOM (Cury et al., 2011; Pikitch et al., 2014; Fifield et al., 2017) to illustrate the usefulness of the products generated in this study beyond the provision of inputs to spatially-explicit ecosystem models. We proceeded in three steps. First, we produced a hotspot map for seabirds from the 0.18° map of relative probability of presence we had constructed from the predictions of the spatial GAM of seabirds. We identified the hotspots of seabirds in the U.S. GOM as those cells where the relative probability of presence of seabirds is equal to or greater than their average relative probability of presence over the entire U.S. GOM (Brodeur et al., 2008, 2014; Grüss et al., 2018a). Second, we fitted a geostatistical binomial GLMM for forage fish (Clupeidae and Exocoetidae; Table 3), which are the main prey of seabirds in the GOM (e.g., Lamb et al., 2016, 2017), and we then produced a hotspot map for forage fish for the U.S. GOM from predictions of the geostatistical GLMM. Lastly, we created maps showing where seabird hotspots and forage fish hotspots overlap in the U.S. GOM.

To construct distribution and hotspot maps for forage fish, we relied on the method developed and applied in Grüss et al. (2018d). This method consists of: (1) extracting presence/absence data from various monitoring datasets integrated in a large monitoring database for fish and invertebrates of the U.S. GOM, along with auxiliary information (year of data collection, geographic coordinates where data were collected); (2) fitting a geostatistical binomial GLMM to these presence/absence data, which integrates the fixed effect of year and the random effect of monitoring program; (3) generating a 0.18° map of probability of presence in the U.S. GOM from the predictions of the fitted geostatistical GLMM; and (4) identifying fish hotspots in the U.S. GOM, as those cells where the probability of presence of fish is equal to or greater than their average probability of presence over the entire U.S.

Table 3
Forage fish of the Gulf of Mexico.

Family	Species
Clupeidae	Alewives (<i>Alosa</i> spp.), finescale menhaden (<i>Brevoortia gunteri</i>), Gulf menhaden (<i>Brevoortia patronus</i>), yellowfin menhaden (<i>Brevoortia smithi</i>), Atlantic menhaden (<i>Brevoortia tyrannus</i>), false herring (<i>Harengula clupeiola</i>), redear herring (<i>Harengula humeralis</i>), scaled sardine (<i>Harengula jaguana</i>), Atlantic thread herring (<i>Opisthonema oglinum</i>), Spanish sardine (<i>Sardinella aurita</i>), Brazilian sardinella (<i>Sardinella brasiliensis</i>)
Exocoetidae	Bearded flyingfish (<i>Cheilopogon cyanopterus</i>), bandwing flyingfish (<i>Cheilopogon exsiliens</i>), doublebearded flyingfish (<i>Cheilopogon furcatus</i>), Atlantic flyingfish (<i>Cheilopogon heterurus</i>), clearwing flyingfish (<i>Cypselurus comatus</i>), blunt-snouted flying fish (<i>Exocoetus obtusirostris</i>), flying fish (<i>Hirundichthys affinis</i>), blackwing flyingfish (<i>Hirundichthys rondeletii</i>), sailfin flyingfish (<i>Paraxocoetus brachypterus</i>)

GOM. This method relies on the R package “VAST”, which is publicly available online (Thorson et al., 2015). The reader is referred to Grüss et al. (2018d) and Appendix A5 for more details.

Next, we identified the 0.18° cells in the U.S. GOM that are both seabird hotspots and forage fish hotspots (Grüss et al., 2018d). This allowed us to create a 0.18° map showing where hotspots of seabirds and forage fish overlap in the U.S. GOM. This map could serve as the basis of a hypothetical MPA planning scenario evaluated with spatially-explicit simulation models (Grüss, 2014; Ainsworth et al., 2015) or applications of conservation planning software packages such as Marxan or Marxan with zones (Watts et al., 2009; Metcalfe et al., 2015), where fishing would not be allowed in some of the 0.18° cells where the hotspots of seabirds and forage fish overlap so as to protect sufficient food resources for seabirds in the U.S. GOM (Cury et al., 2011; Pikitch et al., 2014; Fifield et al., 2017). In practice, since closing large areas to fishing can be prohibitively costly (Grüss, 2014), only some of the cells where hotspots of seabirds and forage fish overlap should be selected to build a proposed MPA scenario, based on a number of criteria (e.g., local biodiversity, distance from fishing ports).

For illustration purposes, we also calculated the number of 0.18° cells that are hotspots for both seabirds and forage fish in each polygon of the Atlantis-GOM ecosystem model. The resulting map showing an index of spatial overlap between hotspots of seabirds and forage fish could serve as the basis of a hypothetical MPA planning scenario explored with the Atlantis-GOM model.

3. Results

3.1. Seabird sightings

The DIVER survey provided us with 7067 sightings for diving birds, 25,945 for surface-feeding birds, and 28,219 for seabirds. The monthly number of sightings ranged from 111 to 1801 for diving birds (mean = 883 sightings), from 677 to 6716 for surface-feeding birds (mean = 3243 sightings), and from 681 to 7129 for seabirds (mean = 3527 sightings). The bottom depths associated with sightings tended to be larger for surface-feeding birds and seabirds (range = 0–631 m; mean = 6 m) than for diving birds (range = 0–328 m; mean = 5 m). The three species groups considered in this study (i.e., diving birds, surface-feeding birds, and seabirds) were all observed by the DIVER survey over the entire U.S. GOM (Fig. 4b–d).

3.2. GAM fitting and performance

Pearson's correlation coefficient between distance from shore and SST was greater than 0.7 for all species groups (i.e., diving birds, surface-feeding birds, and seabirds). Moreover, preliminary analyses revealed that distance from shore explained a greater proportion of the deviance in relative probability of presence than SST for all species groups. Therefore, SST was excluded from all the non-spatial and spatial GAMs fitted in the present study. Pearson's correlation coefficient between bottom depth and distance from shore was also greater than 0.7 in the case of diving birds, leading to the exclusion of distance from shore from the non-spatial GAM of the species group. Moreover, the collinearity analysis suggested discarding distance from shore and

precipitation from the spatial GAM of seabirds, because Pearson's correlation coefficient between northings and distance from shore was equal to -0.71 and Pearson's correlation coefficient between northings and precipitation was equal to 0.85. Thus, five of the eight original environmental predictors were retained in the spatial GAM of seabirds: bottom depth, surface chlorophyll-a concentration, wind speed, oceanic current speed, and SSH.

All fitted GAMs explained a relatively low proportion of the deviance in the data. The non-spatial GAMs of diving birds and surface-feeding birds explained 25% and 31.1% of the deviance in the data, respectively. The spatial GAM of seabirds explained 32.4% of the deviance in the data. Moreover, the examination of empirical variograms revealed that the residuals from all fitted GAMs showed significant spatial autocorrelation (Fig. A6).

The median AUCs of the GAMs of diving birds, surface-feeding birds and seabirds were equal to 0.796, 0.828 and 0.845, respectively. Thus, the median AUCs of all the GAMs fitted in this study were all greater than 0.7. Therefore, all GAMs passed the validation test. Moreover, all fitted GAMs appeared to be reasonably calibrated, as the slope of the calibration plot was close to 1 for all species groups (diving birds: 0.966; surface-feeding birds: 0.971; seabirds: 0.955), and as the intercept of the calibration plot was near 0 for all species groups (diving birds: 0.002; surface-feeding birds: 0.002; seabirds: 0.004).

3.3. Distribution maps for diving birds and surface-feeding birds

The non-spatial GAM of diving birds predicted the species group to be primarily present in nearshore waters of the GOM LME (Fig. 5a–b). In the U.S. GOM, the highest relative probabilities of presence of diving birds were found off the Florida Keys, along the southwestern Florida coastline, and in the coastal areas of Mississippi, eastern Louisiana and southern Texas. In the southern GOM, the highest relative probabilities of presence of diving birds were found in Tamaulipas coastal waters and on the Campeche Bank (Fig. 5a–b). Precipitation was the main parameter explaining the relative probability of presence of diving birds, followed by bottom depth (Fig. 6a). The relative probability of presence of diving birds was predicted to decrease with precipitation when precipitation was above 70 mm (Fig. 7). Precipitation is relatively low all year round in Tamaulipas coastal waters and on the Campeche Bank (Fig. A2), which may explain the high relative probabilities of presence predicted for diving birds in these regions. The relative probability of presence of diving birds was also predicted to decrease with increasing bottom depth when the environmental predictor ranged between 0 and 125 m, and to increase with bottom depth when bottom depth was larger than 125 m. However, the effect of bottom depth on diving bird relative probability of presence was negative when bottom depth was greater than ca. 25 m (Fig. 7).

The non-spatial GAM of surface-feeding birds predicted the species group to be mostly present on the Campeche Bank, in northern Veracruz and Tamaulipas coastal waters, in the coastal areas of Mississippi, eastern Louisiana and Texas, and in the region of West Florida between Sarasota and Naples (Fig. 5c–d). Distance from shore, precipitation, and surface chlorophyll-a concentration were the main parameters explaining the relative probability of presence of surface-feeding birds (Fig. 6b). The relative probability of presence of surface-

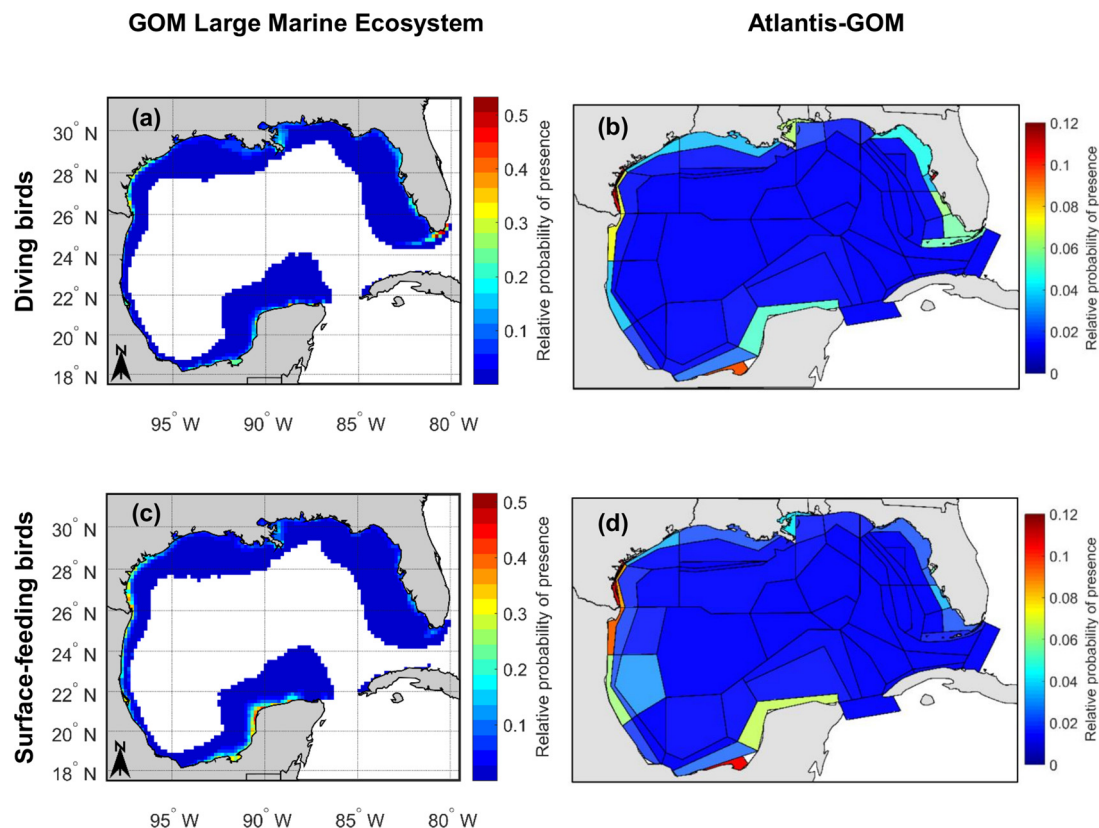


Fig. 5. Maps produced for (a–b) diving birds and (c–d) surface-feeding birds. (a, c) are maps of relative probability of presence for the entire Gulf of Mexico (GOM) Large Marine Ecosystem generated from the predictions of non-spatial generalized additive models (i.e., not integrating an interaction term between eastings and northings), while (b, d) are maps of relative probability of presence usable in the Atlantis-GOM ecosystem model (such that the sum of relative probability of presence for each map is equal to 1.0).

feeding birds tended to decrease with distance from shore, and the effect of distance from shore on surface-feeding bird relative probability of presence was negative when distance from shore was greater than 50 m (Fig. 8). The relative probability of presence of surface-feeding bird was also predicted to decrease with increasing precipitation. Moreover, surface chlorophyll-a concentration had a positive effect on the relative probability of presence of surface-feeding birds when the environmental predictor ranged between 3 and 19 $\text{mg}\cdot\text{m}^{-3}$ (Fig. 8). Precipitation is relatively low in northern Veracruz and Tamaulipas coastal waters and on the Campeche Bank, while surface chlorophyll-a concentration is usually optimal for surface-feeding bird presence ($3\text{--}19\text{ mg}\cdot\text{m}^{-3}$) in these regions all year round (Fig. A2), which may explain the high relative probabilities of presence predicted for surface-

feeding birds in these regions.

3.4. Preference functions for the seabird group represented in WFS Reef fish Ecospace

We produced five preference functions for seabirds for the WFS Reef fish Ecospace ecosystem model based on the outcomes of the collinearity analysis: one for bottom depth, one for surface chlorophyll-a concentration, one for wind speed, one for oceanic current speed, and one for SSH (Fig. 9). Bottom depth had a much larger effect on relative probability of presence of seabirds than the other four environmental parameters. The relative probability of presence of seabirds was predicted to decrease sharply with increasing bottom depth, and seabird

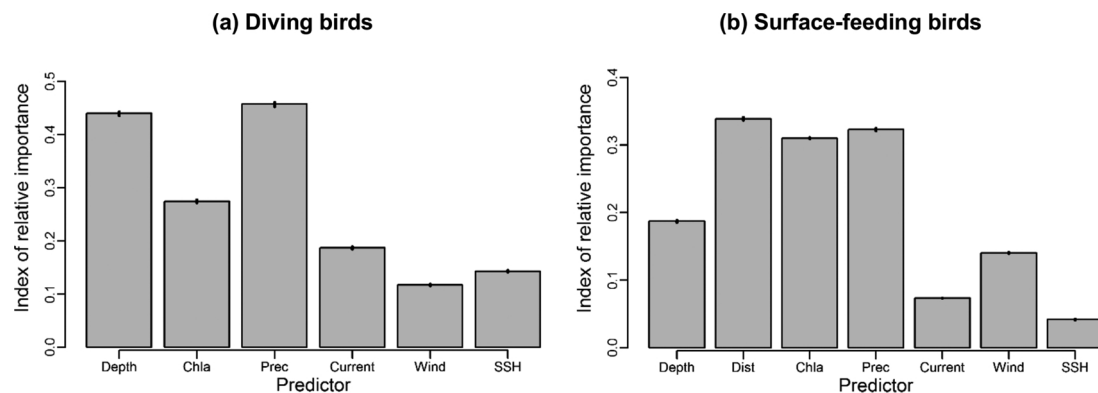


Fig. 6. Relative importance of predictors in the spatial patterns of relative probability of presence of (a) diving birds and (b) surface-feeding birds predicted by binomial generalized additive models. Vertical black lines indicate confidence intervals. Depth = bottom depth – Dist = distance from shore – Chla = surface chlorophyll-a concentration – Prec = precipitation – Current = oceanic current speed – Wind = wind speed – SSH = sea surface height.

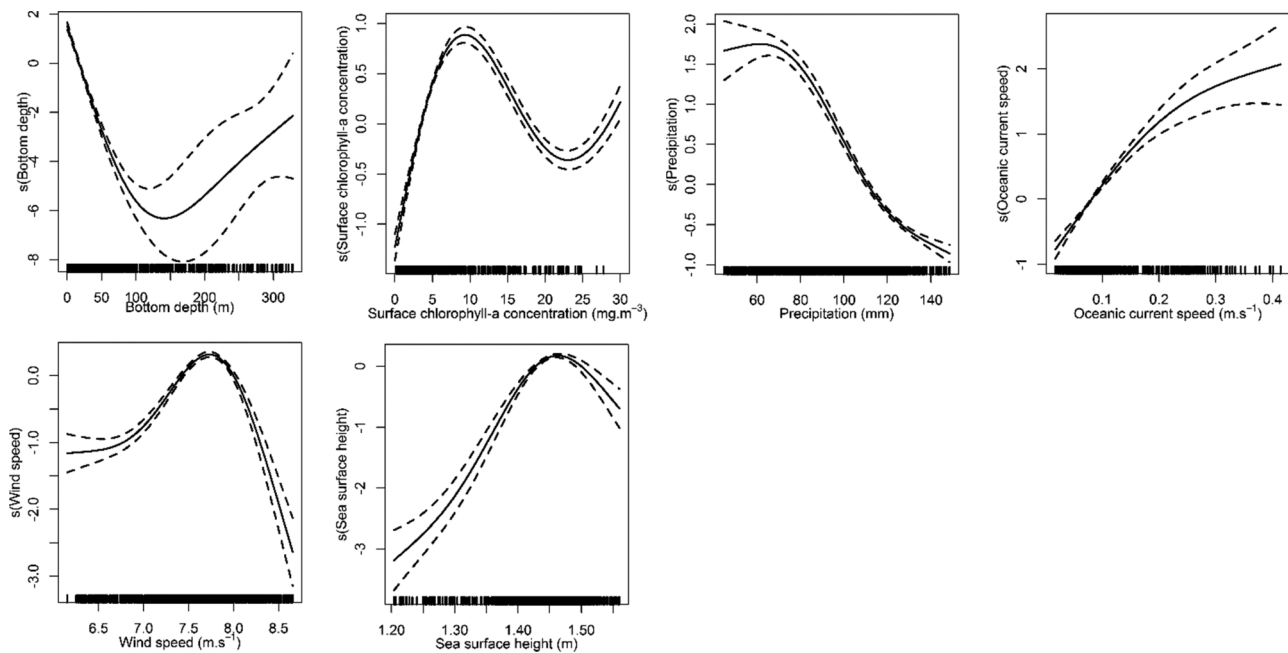


Fig. 7. Smoothed curve of the additive effect to the estimated relative probability of presence of diving birds for the individual environmental parameters considered in the generalized additive model of the species group. Dotted lines represent 95% confidence intervals, and each mark along the x-axis a single observation. Note that the scale of the y-axis differs from one panel to the next for display purposes.

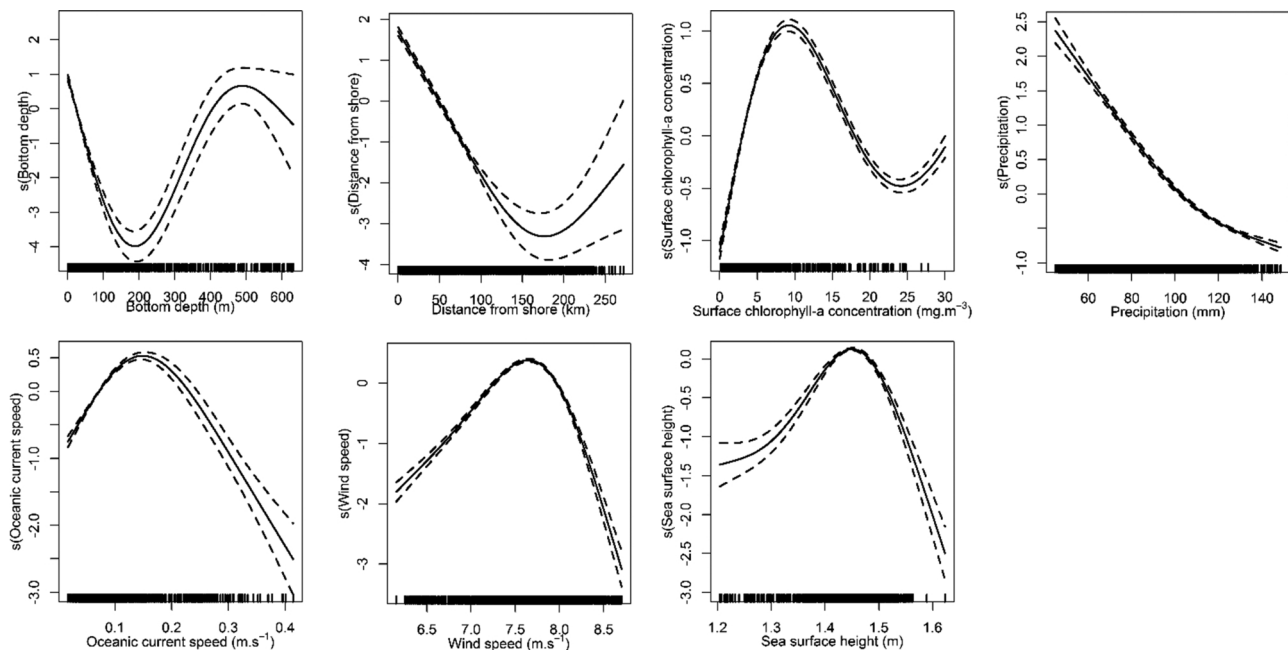


Fig. 8. Smoothed curve of the additive effect to the estimated relative probability of presence of surface-feeding birds for the individual environmental parameters considered in the generalized additive model of the species group. Dotted lines represent 95% confidence intervals, and each mark along the x-axis a single observation. Note that the scale of the y-axis differs from one panel to the next for display purposes.

preference for bottom depths greater than ca. 75 m was close to 0 (Fig. 9a). Oceanic current speed had a larger effect on seabird relative probability of presence than surface chlorophyll-a concentration, wind speed and SSH (Fig. 9b–e). The relative probability of presence of seabirds was predicted to decrease with increasing oceanic current speed when oceanic current speed was lower than 0.1 m.s^{-1} , to increase with increasing oceanic current speed when the environmental predictor ranged between 0.1 and 0.25 m.s^{-1} , and to decrease with increasing oceanic current speed above 0.25 m.s^{-1} (Fig. 9d).

3.5. Provision of a basis for a hypothetical MPA planning scenario

The maps produced from the predictions of the spatial GAM of seabirds indicates that seabirds have a high relative probability of presence in the coastal waters of Texas, Louisiana, Mississippi and southwestern Florida and, to a lesser extent, in Apalachee Bay (Fig. 10a–b). After the geostatistical GLMM of forage fish was fitted and validated (Appendix A5), predictions were made with the model to map the spatial distribution and hotspots of forage fish in the U.S. GOM (Fig. 10c–d) and, subsequently, to generate maps showing the areas of

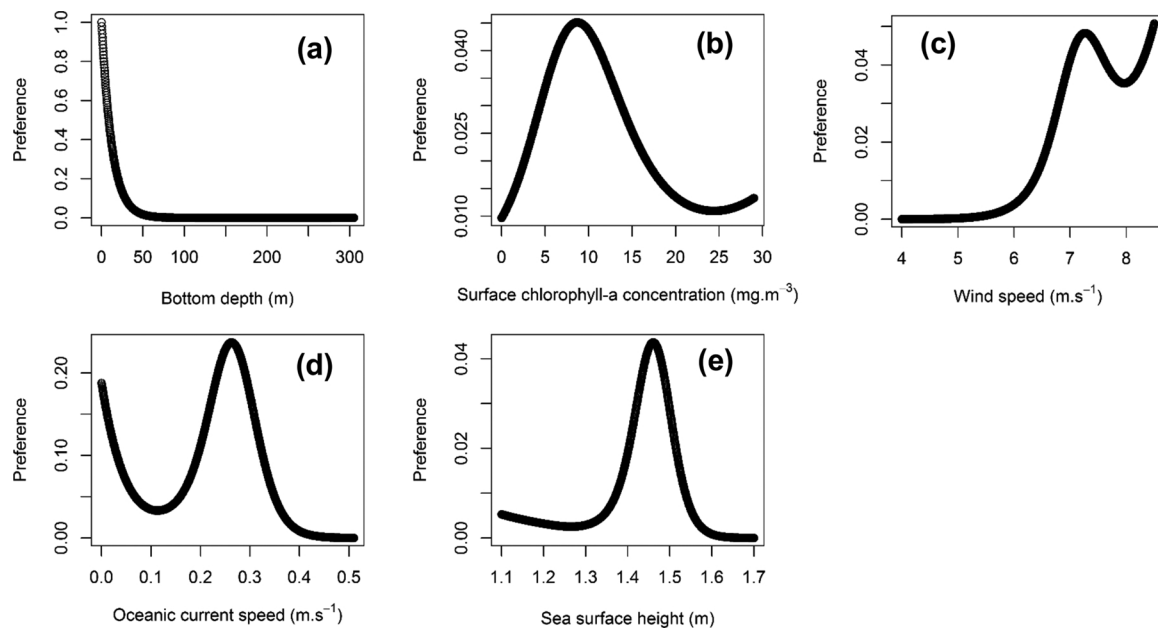


Fig. 9. Preference functions produced for the seabird group represented in the WFS Reef fish Ecospace ecosystem model. Note the different scales on the y-axes.

the U.S. GOM that are both seabird hotspots and forage fish hotspots (Fig. 10e–f). The latter maps suggest that hypothetical MPAs aiming to protect forage fish for seabirds in the U.S. GOM should be implemented primarily within the coastal regions of the Louisiana-Texas shelf, within the coastal region of West Florida located between Sarasota and Naples and/or within Apalachee Bay (Fig. 10e–f).

4. Discussion

We used a method for producing distribution maps and preference functions for spatially-explicit ecosystem models from presence-only data. This method was needed because recent methods for representing species distributions in spatially-explicit ecosystem models relied on presence/absence data (Grüss et al., 2018a,b,d), yet presence-only data are more readily available than presence/absence data for many species such as seabirds (Sullivan et al., 2009; Wood et al., 2011).

We demonstrated the method by applying it to the seabird groups represented in two ecosystem models of the GOM (Atlantis-GOM and WFS Reef fish Ecospace). It could be applied to other marine organisms explicitly considered in spatially-explicit models of the GOM and other marine systems. In the GOM, the method employed here could also be useful for constructing distribution maps for West Indian manatee for the Atlantis-GOM model from opportunistic monitoring data. However, it would be necessary to first enhance the method to address the issue of sampling bias that usually affects opportunistic monitoring data (see below). Our method could also be particularly beneficial to marine regions where monitoring data are scarce even for many fish and invertebrate species, and which rely on presence-only data, including online data sources such as Ocean Biogeographic Information System (OBIS) data (e.g., the Gulf of Gabès in Tunisia; Hattab et al., 2013, 2014).

The critical step of our method is the generation of pseudo-absences (Step 2 in Fig. 3). Several procedures have been proposed in the literature for generating pseudo-absences. Several authors (Pearce and Boyce, 2006; Aarts et al., 2012; Renner and Warton, 2013; Renner et al., 2015) have suggested that absences be selected within the spatial extent of the region of interest either at random (as we did) or on a regular mesh. We generated absences independently for each month to produce absences as objectively as possible since the area sampled varies across months. Other authors (e.g., Hengl et al., 2009; Hattab et al., 2013, 2014) have produced absences for SDMs using habitat

suitability maps predicted by the Ecological Niche Factor Analysis (ENFA) and the geographical coordinates of presence-only records. However, generating absences conditional on the presence-only data and without reference to *a priori* assumptions about sampling effort may result in significant bias, because this approach does not constrain that both presence and pseudo-absence data be random samples from the defined spatial boundaries of the study area (Renner et al., 2015).

We acknowledge two limitations of our method: the presence of spatial structure in GAM residuals, and the fact that our method does not account for detectability. First, the residuals of our non-spatial GAMs developed for extrapolation (not integrating an interaction term between eastings and northings), but also the residuals of the spatial GAM of seabirds (integrating an interaction term between eastings and northings), showed significant spatial autocorrelation (Fig. A6). This yields uncertainties on the statistical interpretation of significance for environmental covariates. Therefore, we recommend that future studies that do not need to perform extrapolations over large marine regions implement our method with geostatistical models accounting for spatial variation in relative probability of presence residuals in lieu of GAMs. One way forward may be to include non-linearities in the relationships between environmental predictors and marine organisms in the geostatistical models implemented using the R package “VAST”, fit the enhanced geostatistical models to presence/pseudo-absence data, and make predictions with the enhanced geostatistical models to construct distribution maps and preference functions (Grüss et al., 2017). Second, as is commonly done in the statistical literature, we talked about “presences” in this study, while we should have employed the term “encounters”. In reality, the DIVER low-level aerial bird survey may not have detected some seabird species/groups in some locations even though the species/groups were actually present, due to seabird body size, color, behavior, flight height, Beaufort sea state, cloud cover, light conditions, or distance from the aircraft, among other factors (Buckland et al., 2001; Gjerdrum et al., 2012; Bolduc and Fifield, 2017). Not accounting for species detectability may result in the prediction of biased relative probabilities of presence by GAMs (Monk, 2014). This bias could be reduced in future studies through the use of the two-stage modeling approach developed by Dorazio (2014), which relies on presence-only data for estimating detectability and then on presence/pseudo-absence data for estimating relative probabilities of presence.

Future studies could apply the method we developed to opportunistic monitoring data, which are increasingly available worldwide

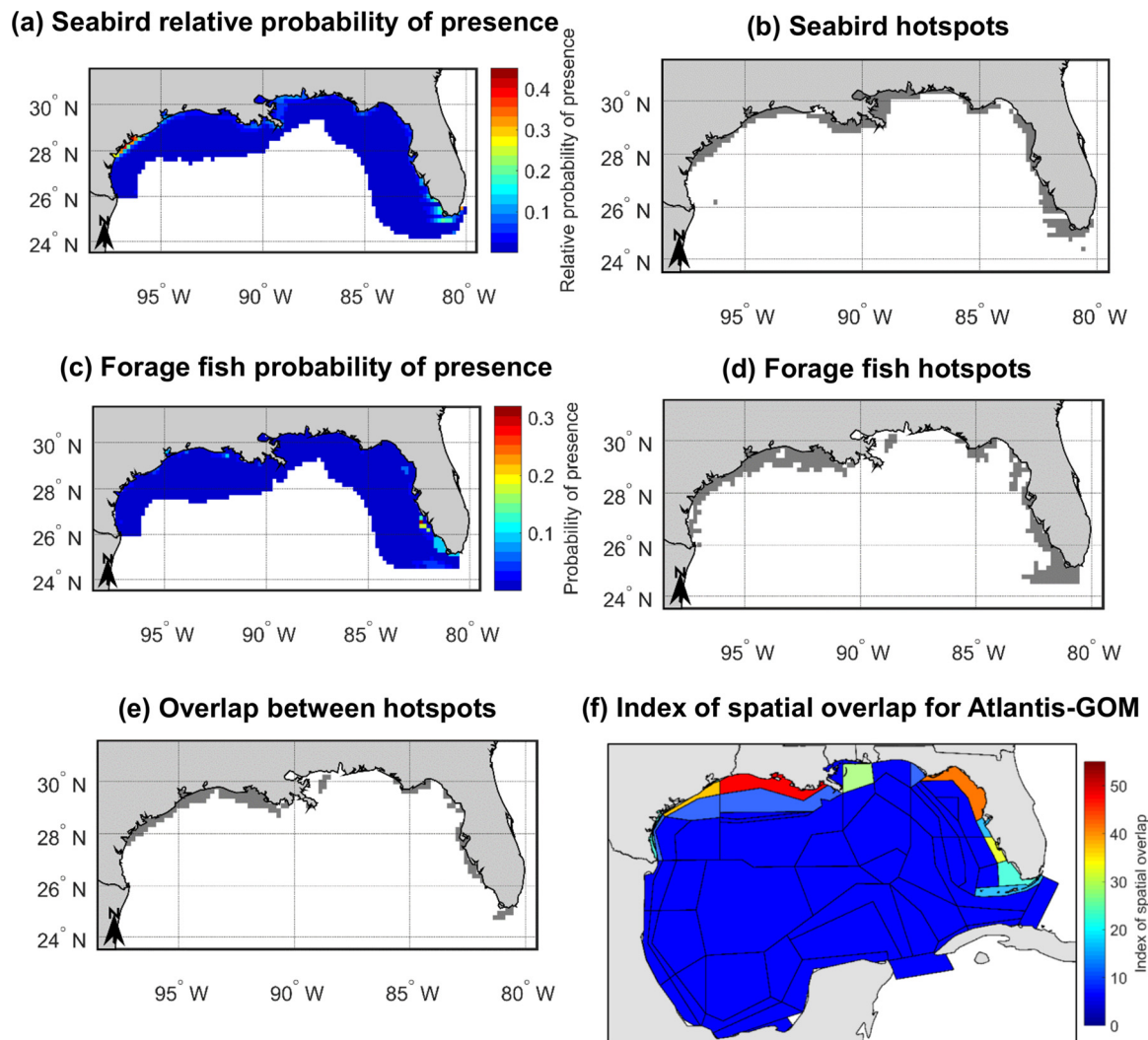


Fig. 10. Provision of a basis for hypothetical marine protected area (MPA) planning scenarios aiming to secure food for seabirds in the U.S. Gulf of Mexico (GOM). (a–b) Production of distribution and hotspot maps for seabirds from the predictions of the binomial generalized additive model of the species group. (c–d) Production of distribution and hotspot maps for forage fish (including Clupeidae and Exocoetidae) from the prediction of a geostatistical generalized linear mixed model developed for them. (e) Construction of a map showing the areas of the U.S. GOM where the hotspots of seabirds and forage fish overlap, which can serve as a basis for a hypothetical MPA planning scenario (e.g., to be explored with a spatially-explicit simulation model). (f) Construction of an “index of spatial overlap” for the Atlantis-GOM ecosystem model, produced by calculating the number of cells of the previous map that are hotspots for both seabirds and forage fish in each Atlantis-GOM polygon; the resulting map can serve as a basis for a hypothetical MPA planning scenario explored with the Atlantis-GOM model.

(Sullivan et al., 2009; Devictor et al., 2010; Dickinson et al., 2010; Wood et al., 2011). However, doing so would require modifications in our modeling framework to account for sampling bias (Williams et al., 2002; Renner and Warton, 2013; Renner et al., 2015). Indeed, with respect to opportunistic monitoring programs, the likelihood of a species being detected and reported as present is often related to sampling characteristics, such as the degree of accessibility to locations related to their distance from populated areas or roads in the case of inland opportunistic seabird surveys (Wiersma, 2010; Callaghan and Gawlik, 2015). In the case of West Indian manatee, the likelihood of a species being reported as present by citizen scientists is influenced by distance to developed areas and distance to boat ramps (Aven et al., 2015; Hieb et al., 2017). If sampling biases existed, additional variables related to properties of the observers should be added to the right side of Eqs. (1) and (2) to model the sampling bias and allow for the extraction of unbiased estimates of relative probability of presence (Renner and Warton, 2013; Renner et al., 2015).

We employed only one seabird monitoring dataset (the DIVER low-level aerial bird survey dataset), although several other seabird monitoring programs have been conducted in the U.S. GOM in the past (e.g.,

Fritts et al., 1983; Ribic et al., 1997; Davis et al., 2000). However, these other monitoring programs have low spatial and/or temporal resolutions that prevented their use in the present study (Love et al., 2015). For example, the NOAA GOM Cetacean II (GulfCet II) survey, which primarily targeted cetaceans, but also monitored seabirds to identify habitat associations between seabirds and cetaceans, took place only in October 1996 and August 1997 in the offshore areas of the north-central GOM (Davis et al., 2000). In addition to their limited spatio-temporal resolution, the other seabird monitoring datasets are difficult to both locate and access (Love et al., 2015). A large monitoring program focusing on seabirds, sea turtles and marine mammals called the “Gulf of Mexico Marine Assessment Program for Protected Species” or “GoMMAPPS” was initiated in 2016 (GoMMAPPS, 2018); fieldwork for GoMMAPPS started in 2017 and will end in 2020. GoMMAPPS covers the entire offshore region of the U.S. GOM out to the U.S. Exclusive Economic Zone and will provide invaluable data which one could utilize in conjunction with the DIVER low-level aerial bird survey data to get a more accurate depiction of the spatial distribution of seabirds in the GOM. We suspect that employing a blending of the DIVER low-level aerial bird and GoMMAPPS data would allow us to develop GAMs

Table 4

Comparison of the insights from the present study and from the literature into the spatial distribution patterns of the diving bird and surface-feeding bird groups represented in the Atlantis-GOM ecosystem model.

Species group	Insights from the present study	Insights from the literature
Diving birds	• In the U.S. Gulf of Mexico (GOM), the highest relative probabilities of presence of diving birds are found off the Florida Keys, along the southwestern Florida coastline, and in the coastal areas of Mississippi, eastern Louisiana and southern Texas. • In the southern GOM, the highest relative probabilities of presence of diving birds are found in Tamaulipas coastal waters and on the Campeche Bank.	• The species making up the diving bird group represented in Atlantis-GOM tend to occupy the entire coast of the U.S. GOM. Thus, brown pelican (<i>Pelecanus occidentalis</i>) is found all along the U.S. coastline, particularly in the nearshore areas of Florida, eastern Louisiana and Texas, and brown pelicans do not expand their foraging beyond 30 kilometers from land (Shields, 2002). The literature indicates that northern gannet (<i>Morus bassanus</i>) is present along the entire U.S. GOM coast, but is more commonly found in nearshore areas of Alabama, in the region between Tampa, Florida and Naples, as well as off the Florida Keys (Mowbray, 2002; National Audubon Society, 2010). Finally, according to the literature, the highest probabilities of presence of common loon (<i>Gavia immer</i>) are found in coastal areas along the Barataria and Vermilion Bays in Louisiana, the Mississippi and Alabama States and the Florida Panhandle, as well as in Apalachee and Tampa Bays (Evers, 2004). • Very little information is available regarding the spatial distribution patterns of the species making up the diving bird group in the southern GOM. Very few data are available for brown pelican nesting colonies in Mexico. These data suggest that, in the Mexican GOM, brown pelican is mostly found in the coastal areas of northern Veracruz and the Campeche Bank (Ridgely et al., 2007). Few or no relevant count data exist for the Mexican and Cuban GOM for northern gannet, and the limited available data suggest that, in the southern GOM, northern gannet is essentially present in the coastal areas of Tamaulipas State (Ridgely et al., 2007; National Audubon Society, 2010). Few or no relevant count data exist for the Mexican and Cuban GOM for common loon, and the limited available data suggest that, in the southern GOM, common loon is essentially present in the coastal areas of Tamaulipas State and northern Veracruz (National Audubon Society, 2010).
Surface-feeding birds	• In the U.S. GOM, surface-feeding birds are mostly present in the coastal areas of Mississippi, eastern Louisiana and Texas, and in the region of West Florida between Sarasota and Naples. • In the southern GOM, surface-feeding birds are mostly present on the Campeche Bank, in northern Veracruz and Tamaulipas coastal waters.	• The species making up the surface-feeding bird group represented in Atlantis-GOM tend to occupy the entire coast of the U.S. GOM. The probability of presence of black skimmer (<i>Rynchops niger</i>) is high all along the U.S. GOM coastline, particularly off Naples and in Tampa and Apalachicola Bays in Florida as well as in the coastal areas of Mississippi, eastern Louisiana and Texas (Cornwell University, 2011). The probability of presence of least tern (<i>Sterna antillarum</i>) is also high all along the U.S. GOM coastline (Thompson et al., 1997; Cornwell University, 2011). The probability of presence of royal tern (<i>Thalasseus maximus</i>) is highest in the nearshore areas of the U.S. GOM located between the Mexican border and the Mississippi-Alabama border and between Apalachicola, Florida and Sarasota (Buckley and Buckley, 2002). • Very limited information is available regarding the spatial distribution patterns of surface-feeding bird species in the southern GOM. Black skimmer was reported in the coastal areas of Tamaulipas State, northern Veracruz State and the Campeche Bank and, to a lesser extent, on the coast of Cuba in winter (Gochfeld and Burger, 1994; Ridgely et al., 2007). Very few data are available for least tern and royal tern nesting colonies in Mexico. These data suggest that, in the Mexican GOM, the probability of presence of least tern is highest on the Tamaulipas coast (Zdravkovic, 2006), while that of royal tern is highest in the inshore and offshore waters of Campeche Bank (Tunnell and Chapman, 2000).

explaining a larger proportion of the deviance in the data (which was relatively low in this study) and to determine more significant and more reliable relationships between seabirds and environmental parameters.

The distribution maps produced for the diving and surface-feeding bird groups represented in Atlantis-GOM in this study (Fig. 5) are, to our knowledge, the first maps produced for seabirds from SDM predictions at the scale of the entire GOM LME. The spatial distribution patterns predicted by our maps tend to be supported by the existing literature (Table 4). However, it is difficult to gauge the relative probabilities of presence predicted by our GAMs for the southern GOM, due to a lack of information on the spatial distribution patterns of seabird species in the Mexican GOM and, to a larger extent, in the Cuban GOM (Table 4). We, therefore, encourage future studies to employ the GoMMAPPS data (when they are released) and/or, ideally, monitoring data for the southern GOM if those can be accessed, to re-examine our results for the southern GOM. Moreover, to further our knowledge on seabirds, future studies relying on the large amount of data provided by GoMMAPPS should explore the spatial and seasonal

distribution patterns of individual seabird species of the GOM. While ecosystem models simulate the dynamics of broad species groups for computational convenience, the spatio-temporal distribution patterns of GOM seabirds can vary substantially among species (Duncan and Havard, 1980; Clapp et al., 1982a,b, 1983; Burger, 2017).

The products generated in this study are not useful solely for representing species distributions in spatially-explicit ecosystem models from presence-only data. To demonstrate this, we combined the hotspot map of seabirds for the U.S. GOM with a hotspot map produced for forage fish to provide a basis for a hypothetical MPA planning scenario aiming to secure food for seabirds (Fig. 10). Our additional analysis suggested that, to protect forage fish for seabirds in the U.S. GOM, hypothetical MPAs should be implemented primarily within the coastal regions of the Louisiana-Texas shelf, within the coastal region of West Florida located between Sarasota and Naples and/or within Apalachee Bay (Fig. 10e–f). Another potential use of the products generated in this study is the identification of bycatch areas (e.g., the areas of the U.S. GOM where seabirds are caught as bycatch in the pelagic longline

fishery) to allow for the design of bycatch mitigation strategies. We hope that the method presented in this paper will benefit ecosystem modeling and other efforts in the GOM and elsewhere that consider species for which presence-only rather than presence-absence data are available.

Authorship statement

AG designed and analyzed the models; AG and MDD conceived the models; AG, MDD and EC compiled the monitoring dataset and environmental database; AG, MDD, EC, CHA, MSL, JSG, JMT and EAB wrote the paper; all authors contributed to manuscript revision.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.fishres.2018.10.011>.

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